

Note 4.3 Similarity

Intro

In this section, we will consider different matrix representations of linear operators and characterize the relationship between matrices representing the same linear operator.

Theorem 4.3.1

Let $E = \{\mathbf{v}_1, \dots, \mathbf{v}_n\}$ and $F = \{\mathbf{w}_1, \dots, \mathbf{w}_n\}$ be two **ordered bases** for a vector space V and let L be a linear operator on V . Let S be the transition matrix representing the changing from F to E . If A is the matrix representing L with respect to E , and B is the matrix representing L with respect to F , then $B = S^{-1}AS$.

Understanding (feat. 3b1b)

This process is first use S to transform the basis from F to E , that is, transform the representation of a vector from F to E . Then we use A to do this linear transformation and AS is the result but with the representation of E . And finally we just need to transform the representation of the result vector from E to F . And the process is the representing L with respect to F .

Proof

Let

$$\mathbf{y} = S\mathbf{x}, \mathbf{t} = A\mathbf{y}, \mathbf{z} = B\mathbf{x}$$

And we can suppose the vector $\mathbf{v}$ can be represented as

$$\begin{aligned}\mathbf{v} &= x_1\mathbf{w}_1 + x_2\mathbf{w}_2 + \dots + x_n\mathbf{w}_n \\ \mathbf{v} &= y_1\mathbf{v}_1 + y_2\mathbf{v}_2 + \dots + y_n\mathbf{v}_n\end{aligned}$$

Since A represents L with respect to E , and B represents L with respect to F , we have

$$\mathbf{t} = [L(\mathbf{v})]_E, \mathbf{z} = [L(\mathbf{v})]_F$$

The transition matrix from E to F is S^{-1} thus

$$S^{-1}\mathbf{t} = \mathbf{z}$$

Then we have

$$S^{-1}AS\mathbf{x} = B\mathbf{x}$$

hence $S^{-1}AS = B$.

Definition of Similar

Let A and B be $n \times n$ matrices. B is said to be similar to A if there exists a nonsingular matrix S such that $B = S^{-1}AS$.